

Datasheet

Product: Smart Spot Air Quality & Sound Level Meter

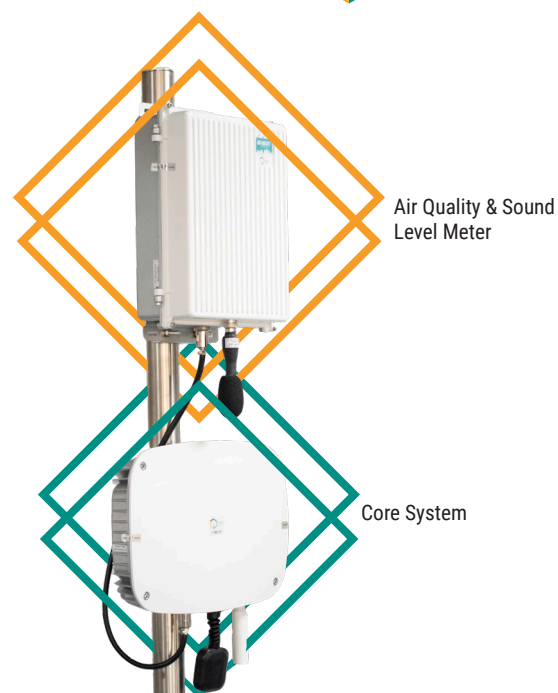


Smart Spot Air Quality & Sound Level Meter

Smart Spot is an advanced and robust IoT device that allows monitoring of different environmental parameters such as air quality (gases and PM), temperature, humidity, flow of people (crowd monitoring) and noise. Therefore, a unique monitoring station is offered with a high capacity to obtain data on the context, environment and ambient to know the evolution of a city.

The inclusion of all these sensors and capacities in the same device provides savings in the installation, maintenance and management of all sensors, as well as in communications.

For the connection possibilities of this device, multiple protocols are supported: Wi-Fi, GSM/GPRS (M2M), LoRa (LPWAN) and 4G/NB-IoT.



CORE SYSTEM

Characteristics

Operating System	Braking System Watchdogs
	Industrial & real-time operating system (FreeRTOS)
	Industrial/hazardous environments ready
CPU	Dual-Core a 240 MHz
Antennas	Multi antenna IP68 anti-vandalism (GPS/M2M/WiFi)
Health device monitoring	Temperature
	Humidity
Vandalism detection	Accelerometer
	Gyroscope

Power Supply

Energy consumption	180-300 mA Active
Voltage (nominal)	5V
Battery	20.000 mA
Solar panel	6.5V

Communications

Networks	WiFi
	LoRa
	GPRS
Protocols	MQTT
	OMA LwM2M
	ETSI NGSI (FIWARE) - CEF Broker
	HTTP/RESTful API
	SENTILO Open Platform

Enclosure

Protection	IP65
	UV
Size	300x220x36,7 cm
Material	Aluminium
Anchoring system	Anti-vandalism security
Weight	1,8 kg

Datasheet

Product: Smart Spot Air Quality & Sound level meter



Extensions

Nocive & greenhouse effect gasses* (Air Quality monitoring)	NO ₂ , H ₂ S, CO, NO, SO ₂ , O ₃ , NH ₃ y CO ₂
Particles Matter (PM)	PM1, PM2.5 y PM10
Ambient sensors	Temperature & humidity
Flow of people (Crowd Monitoring)	WiFi y BLE
Sound Level Meter	Class 2 (40-115 dB)

*Up to 6 gasses

Extensions enclosure

Protection	Aluminium IP65
Weight	2,2 kg
Anchoring system	Anti-vandalism security
Size	100x220x280 mm

Nocive & greenhouse effect gasses (Air Quality)

Core system	Active air flow pump
	Coarse particles filter
	Controller system of Air Quality extension
	Dual-gas board (2, 4 o 6 gases)

Sensor technology

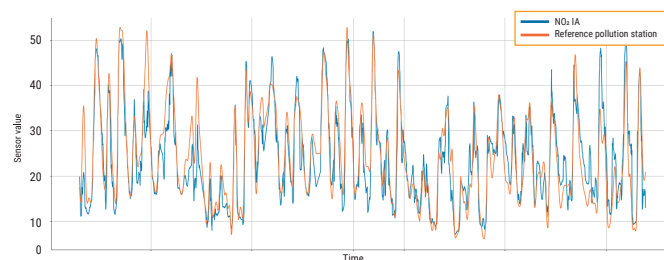
Type of sensor	Electrochemical
Humidity range	[15, 85] % hr
Temperature range	[-20, 40] °C
Lifetime	24 months

Data Quality & Calibrations Service

Calibration equipment	Calibration with reference gas with external composition and stability certification (LINDE)
	Composition & stability external certification
	Equipment UNE-EN ISO/IEC 17025, EPA Agency
Artificial Intelligence models	Drift compensation
	Elimination of atypical data
	Adapted model for sensors precision

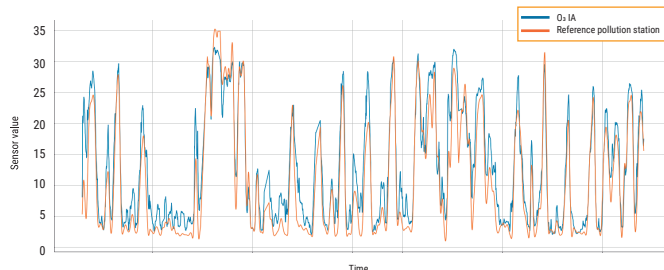
NO₂ (correlation coef. 0,80)

Range	0-5 ppm
Resolution	2 ppb
Exposition maximum range	50 ppm
Cross sensitivity	CL ₂ > H ₂ S> NO> SO ₂ , CO



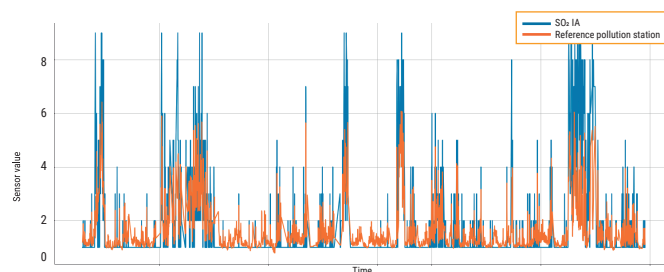
O₃ (correlation coef. 0,89)

Range	0-5 ppm
Resolution	3 ppb
Exposition maximum range	50 ppm
Cross sensitivity	CL ₂ > H ₂ S> NO> SO ₂ , CO



SO₂ (correlation coef. 0,80)

Range	0-5 ppm
Resolution	3 ppb
Exposition maximum range	100 ppm
Cross sensitivity	O ₃ , NO ₂ > CL ₂ > H ₂ S, NO, CO



Datasheet

Product: Smart Spot Air Quality & Sound level meter



CO (correlation coef. 0,80)

Range	0-12 ppm
Resolution	5 ppb
Exposition maximum range	2.000 ppm
Cross sensitivity	H ₂ S> NO ₂ , NO

H₂S

Range	0-5 ppm
Resolution	2 ppb
Exposition maximum range	50 ppm
Cross sensitivity	NO ₂ >SO ₂ > CL ₂ > NO

NO

Range	0-5 ppm
Resolution	10 ppb
Exposition maximum range	50 ppm
Cross sensitivity	H ₂ S>NO ₂ >SO ₂ , Cl ₂

NH₃

Range	0-20 ppm
Resolution	20 ppb
Exposition maximum range	200 ppm
Cross sensitivity	CL ₂ , NO, SO ₂ > H ₂ S; NO ₂ > CO> H ₂

CO₂

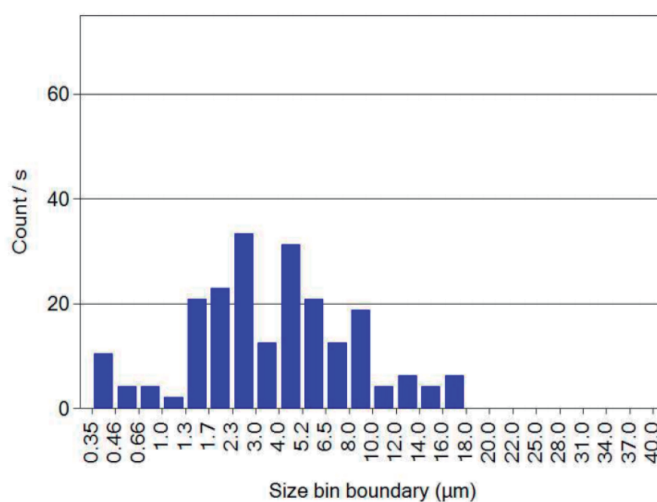
Technology	Optical
	Specific board and housing
Range	0-5.000 ppm
Resolution	1 ppm
Lifetime	24 months

Certificados sensores gases y PM

Calibration certificate		Broadcast with each equipment during the fabrication and calibration in laboratory
Precision traceability certificate	 Centro Tecnológico de la Energía y del Medio Ambiente	With respect to reference laboratory/equipment under the ENAC regulations
Certification IEEE - 2510		Data Quality for IoT sensors.

Particles Matter - PM (Dust & Pollen)

Core system	Controller system of Air Quality extension
	Anti-humidity system
	Air flow pump
Measurement range	0,35 a 40 µm
Particles/second	10,000
Size of particles	PM1, PM2.5 y PM10
Max mass flow rate	2,000 µg/m3



www.hopu.org

+34 868 923 923

Luis Buñuel 6, 30562, Murcia (Spain)

Air Quality Certifications



People flow

Configuration	Independent configuration (WiFi y BLE)
Times range	3 time ranges with simultaneously aggregated
	Aggregated time configurable between 1 m 1 h. (3 ranges)

Hash algorithms	Obfuscated WiFi/BLE identifiers SHA1 and MDS since detection
	Hash algorithm for obfuscation configurable (MDS and SHA1)
	Key to Hash algorithm of configurable obfuscation
	Individual device report detected in Hash format (SHA1 and MDS)
	Collective device reporting detected in Hash format (SHA1 and MDS)

Temperature & humidity

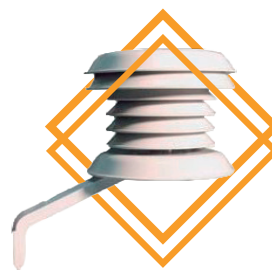
Temperature

Resolution	0.01°C
Precision	±0.1°C
Range	-40°C a +125°C

Humidity

Resolution	0.01 %HR
Precision	±1.5 %HR
Range	0 %HR a 100 %HR

Probe	RS3-B solar radiation protection
-------	----------------------------------



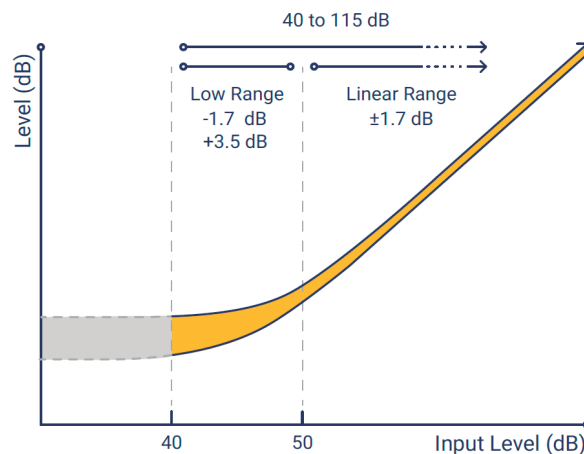
RS3-B solar radiation protection

Sound Level Meter

Characteristics

Operating range SPL	40 - 115 dB
Weighting frequency	Filter IEC 61672-1 A
Weighting time	IEC 61672-1 Slow (S) y Fast (F)
Certification	ROHS2/CE
Additional features	Continuous exposure monitor
	Threshold detection

Range of action (40-115 dB)



Datasheet

Product: Smart Spot Air Quality & Sound level meter

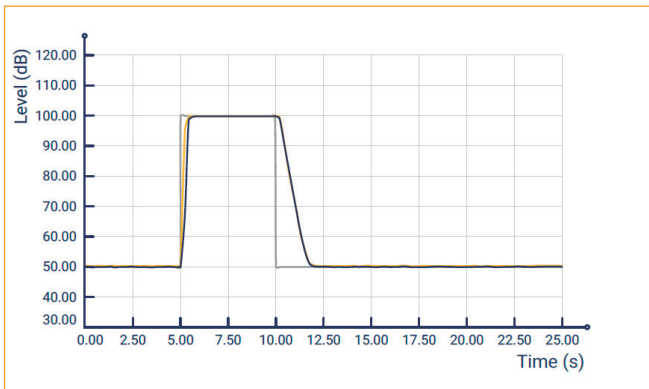


Functions

Available apps	LASFast
	LAFast max
	LAFast min
	LASlow
	LASlow max
	LASlow min
	LAeq

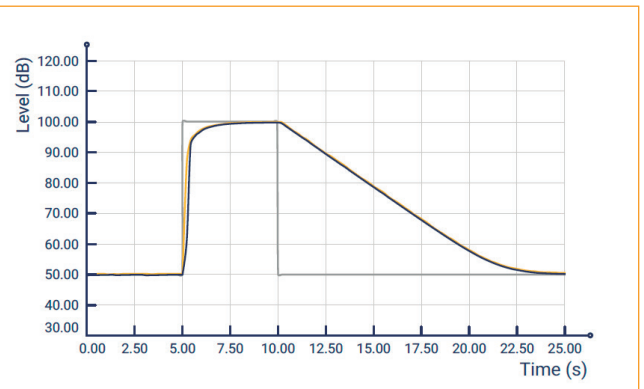


Weighting time - FAST (F)



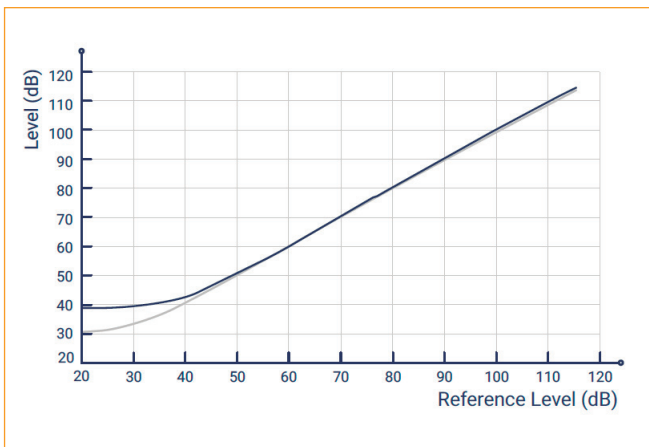
Step response comparison between SPL meter and Class 2 reference device with the fast time constant ($= 0.125s$).

Weighting time - SLOW (S)



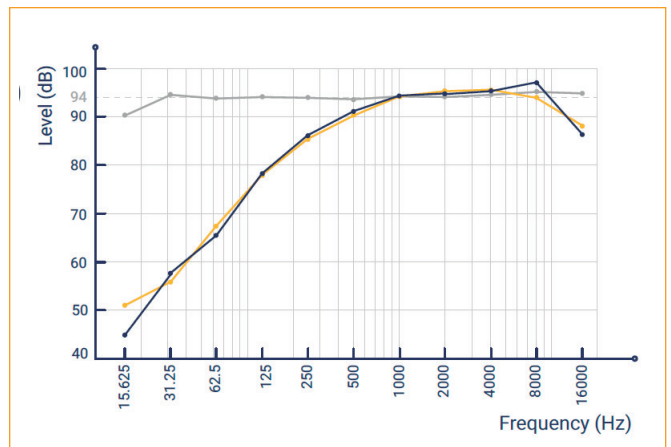
Step response comparison between SPL meter and Class 2 reference device with the slow time constant ($= 1s$).

Range of response (1kHz)



— SPL Meter — Entry level
— Reference*

Response frequency weighting A



*The device used as a reference is a class 2 sound level meter according to IEC 61672 & ANSI S1.4